1.14 Polynomials

Worked Solutions

1 a)

$(x-1)(2x+1)(7x+3) = (2x^2 - x - 1)(7x+3)$

$= 14x^3 + 6x^2 - 7x^2 - 3x - 7x - 3$

$= 14x^3 - x^2 - 10x - 3$

1 b)

$(2x-1)(3x-4)(5x-2)$

$= (6x^2 - 11x + 4)(5x-2)$

$= 30x^3 - 12x^2 - 55x^2 + 20x + 22x - 8$

$= 30x^3 - 67x^2 + 42x - 8$

1 c)

$(4x+3)(2x+1)(3x-2)$

$= (8x^2 + 10x + 3)(3x-2)$

$= 24x^3 - 16x^2 + 30x^2 - 20x + 9x - 6$

$= 24x^3 + 14x^2 - 11x - 6$

2 a)

$8x^3 - 12x^2 - 28x = 4x(2x^2 - 3x - 7)$

2 b)

$10x^4 - 20x^3 - 30x^2 = 10x^2(x^2 - 2x - 3)$

$= 10x^2(x+1)(x-3)$

2 c)

$6x^4 - 45x^3 - 24x^2 = 3x^2(2x^2 - 15x - 8)$

$= 3x^2(2x+1)(x-8)$

3 a)

$2x^2y^3z - 6xy^4z^2 = 2xy^3z(x - 3yz)$

3 b)

$4x^2y^3z^4 + 14x^3y^2z^5 = 2x^2y^2z^4(2y + 7xz)$

3 c)

$3(x-1)^3y^2z - 9(x-1)^2yz^3$

$= 3(x-1)^2yz((x-1)y - 3z^2)$

$= 3(x-1)^2yz(xy - y - 3z^2)$

3 d)

$9x^2(2y-1)^3z - 15x^3(2y-1)z^2$

$= 3x^2(2y-1)z(3(2y-1)^2 - 5xz)$

3 e)

$8x^2(3y-2)^3(z+1) - 4x^3(3y-2)(z+1)^2$

$= 4x^2(3y-2)(z+1)(2(3y-2)^2 - x(z+1))$

4 a)

4

4 b)

5

4 c)

6

5 a)

	x^2	$-3x$	$+1$
x	x^3	$-3x^2$	$+x$
$+2$	$2x^2$	$-6x$	$+2$

$x^3 - 3x^2 + x + 2x^2 - 6x + 2$

$= x^3 - x^2 - 5x + 2$

5 b)

	x^3	$+2x^2$	$-x$	$+3$
$2x$	$2x^4$	$+4x^3$	$-2x^2$	$+6x$
-1	$-x^3$	$-2x^2$	$+x$	$+3$

$2x^4 + 4x^3 - 2x^2 + 6x - x^2 - 2x^2 + x + 3$

$= 2x^4 + 3x^3 - 4x^2 + 7x + 3$

5 c)

	$3x^2$	$-x$	$+2$
x^2	$3x^4$	$-x^3$	$+2x^2$
$+2x$	$+6x^3$	$-2x^2$	$+4x$
$+3$	$+9x^2$	$-3x$	$+6$

$3x^4 - x^3 + 2x^2 + 6x^3 - 2x^2 + 4x + 9x^2 - 3x + 6$

$= 3x^4 + 5x^3 + 9x^2 + x + 6$

5 d)

	$2x^2$	$-4x$	$+1$
x^2	$2x^4$	$-4x^3$	$+x^2$
$+3x$	$+6x^3$	$-12x^2$	$+3x$
-2	$-4x^2$	$+8x$	-2

$2x^4 - 4x^3 + x^2 + 6x^3 - 12x + 3x - 4x^2 + 8x - 2$

$= 2x^4 + 2x^3 - 15x^2 + 11x - 2$

6 a)

$3x^2 + 5x$

6 b)

$\frac{7}{3}x^2 + 3x^9$

6 c)

$x - 5$

6 d)

$(x - 2)^2$

6 e)

$\frac{1}{3x-1}$

6 f)

$\frac{x-5}{(2x+1)(x-5)} = \frac{1}{2x+1}$

6 g)

$\frac{(2x-1)(x-7)}{(2x-1)(x+3)} = \frac{x-7}{x+3}$

7 a)

	$2x^2$	$-7x$	$+3$
$2x$	$4x^3$	$-14x^2$	$+6x$
-4	$-8x^2$	$+28x$	-12

$\frac{4x^3-22x^2+34x-120}{2x-4} = 2x^2 - 7x + 3$

7 b)

	x^3	$+4x^2$	$-x$	$+3$
x	x^4	$+4x^3$	$-x^2$	$+3x$
$+2$	$+2x^3$	$+8x^2$	$-2x$	$+1$

$\frac{x^4+6x^3+7x^2+x+6}{x+2} = x^3 + 4x^2 - x + 3$

7 c)

	$2x^4$	$-x^3$	$+x^2$	$-10x$	$+4$
$3x$	$6x^5$	$-3x^4$	$+3x^3$	$-30x^2$	$+12x$
-6	$-12x^4$	$+6x^3$	$-6x^2$	$+60x$	-24

$\frac{6x^5-15x^4+9x^3-36x^2+72x-24}{3x-6} = 2x^4 - x^3 + x^2 - 10x + 4$

7 d)

	$8x^3$	$-x^2$	$+0x$	$+5$
$4x$	$32x^4$	$-4x^3$	$+0x^2$	$+20x$
-1	$-8x^3$	$+x^2$	$+0x$	-5

$\frac{32x^4-12x^3+x^2+20x-5}{4x-1} = 8x^3 - x^2 + 5$

7 e)

	x^5	$-x^4$	$+0x^3$	$+0x^2$	$+0x$	$+2$
$2x$	$2x^6$	$-2x^5$	$+0x^4$	$+0x^3$	$+0x^2$	$+4x$
$+3$	$+3x^5$	$-3x^4$	$+0x^3$	$+0x^2$	$+0x$	$+6$

$\frac{2x^6+x^5-3x^4+4x+6}{2x+3} = x^5 - x^4 + 2$

7 f)

	x^3	$+2x^2$	$+x$	-5
x^2	x^5	$+2x^4$	$+x^3$	$-5x^2$
$+x$	$+x^4$	$+2x^3$	$+x^2$	$-5x$
$+1$	$+x^3$	$+2x^2$	$+x$	-5

$\frac{x^5+3x^4+4x^3-2x^2-4x-5}{x^2+x+1} = x^3 + 2x^2 + x - 5$

7 g)

	x^5	$-5x^4$	$+0x^3$	$+x^2$	$-8x$	$+12$
x^2	x^7	$-5x^6$	$+0x^5$	$+x^4$	$-8x^3$	$+12x^2$
$+0x$	$+0x^6$	$+0x^5$	$+0x^4$	$+0x^3$	$+0x^2$	$+0x$
$+1$	$+x^5$	$-5x^4$	$+0x^3$	$+x^2$	$-8x$	$+12$

$\frac{x^7-5x^6+x^5-4x^4-8x^3+13x^2-8x+12}{x^2+1}$

$= x^5 - 5x^4 + x^2 - 8x + 12$

8 a)

	x^2	$+8x$	$+11$	
x	x^3	$+8x^2$	$+11x$	-5
$+1$	$+x^2$	$+8x$	$+11$	

$x^2 + 8x + 11$ remainder -5

$\frac{x^3+9x^2+19x+6}{x+1} = x^2 + 8x + 11 - \frac{5}{x+1}$

8 b)

	$2x^2$	$-4x$	-2	
$2x$	$4x^3$	$-8x^2$	$-4x$	-14
-1	$-2x^2$	$+4x$	$+2$	

$2x^2 - 4x - 2$ remainder -14

$\frac{4x^3-10x^2-12}{2x-1} = 2x^2 - 4x - 2 - \frac{14}{2x-1}$

8 c)

	x^3	$+0x^2$	$-x$	$+3$	
x	x^4	$+0x^3$	$-x^2$	$+3x$	$+10$
$+2$	$+2x^3$	$+0x^2$	$-2x$	$+6$	

$x^3 - x + 3$ remainder $+10$

$\frac{x^4+2x^3-x^2+x+16}{x+2} = x^3 - x + 3 + \frac{10}{x+2}$

8 d)

	x^3	$+0x^2$	$-3x$	$+1$	
x^2	x^5	$+0x^4$	$-3x^3$	$+x^2$	$-3x$
$+0x$	$+0x^4$	$+0x^3$	$+0x^2$	$+0x$	$+11$
-1	$-x^3$	$-0x^2$	$+3x$	-1	

$x^3 - 3x + 1$ remainder $-3x + 11$

$\frac{x^3-3x+1}{x^2-1} = x^3 - 3x + 1 + \frac{-3x+11}{x^2-1}$

9 a)

	x^2	$+2x$	$+2$
x	x^3	$+2x^2$	$+2x$
-5	$-5x^2$	$-10x$	-10

$x^3 - 3x^2 - 8x - 10 = (x - 5)(x^2 + 2x + 2)$

9 b)

$x^2 + 2x + 2$ has discriminant $4 - 8 = -4 < 0$

So the only solution to $f(x) = 0$ is $x = 5$

10 a)

	$2x^2$	$-11x$	$+12$
x	$2x^3$	$-11x^2$	$+12x$
$+2$	$+4x^2$	$-22x$	$+24$

$2x^3 - 7x^2 - 10x + 24 = (x + 2)(2x^2 - 11x + 12)$

10 b)

$x^3 - 8x^2 - 14x + 12 = (x + 2)(x - 4)(2x - 3)$

10 c)

$x = -2, x = 4, x = \frac{3}{2}$

10 d)

$f(x - 2) = (x - 2 + 2)(x - 2 - 4)(2(x - 2) - 3)$

$= x(x - 6)(2x - 7)$

$x = 0, x = 6, x = \frac{7}{2}$

11 a)

	x^2	$-x$	$+18$
$2x$	$2x^3$	$-2x^2$	$+36x$
$+1$	$+x^2$	$-x$	$+18$

$2x^3 - x^2 + 35x + 18 = (2x + 1)(x^2 - x + 18)$

11 b)

$x^2 - x + 18$ has discriminant $1 - 4 \times 1 \times 18 = -71 < 0$ so no real solutions to $x^2 - x + 18 = 0$

11 c)

$x = -\frac{1}{2}$

11 d)

$f(x - 4) = (2(x - 4) + 1)((x - 4)^2 - (x - 4) + 18)$

$= (2x - 7)(x^2 - 9x + 38)$

$x = \frac{7}{2}$

12 a)

$f(x) = x(x^2 + 4x - 5) = x(x - 1)(x + 5)$

	x^2	$+x$	-2
x	x^3	$+x^2$	$-2x$
-1	$-x^2$	$-x$	$+2$

$g(x) = (x - 1)(x^2 + x - 2) = (x - 1)(x - 1)(x + 2)$

12 b)

$f(x) - g(x) = x^3 + 4x^2 - 5x - x^3 + 3x - 2$

$= 4x^2 - 2x - 2 = 2(2x^2 - x - 1)$

$= 2(x - 1)(2x + 1)$

So $(x - 1)$ is also a factor of $f(x) - g(x)$

13 a)

Let $f(x) = x^3 - 10x^2 + 13x + 24$

$f(3) = 27 - 90 + 39 + 24 = 0$

$\therefore x - 3$ is a factor of $x^3 - 10x^2 + 13x + 24$

13 b)

Let $f(x) = x^3 - 5x^2 - 10x + 200$

$f(-5) = -125 - 125 + 50 + 200 = 0$

$\therefore x + 5$ is a factor of $x^3 - 5x^2 - 10x + 200$

13 c)

Let $f(x) = 4x^3 - 21x^2 + 29x - 6$

$f(\frac{1}{4}) = \frac{1}{16} - \frac{21}{16} + \frac{29}{4} - 6 = 0$

$\therefore 4x-1$ is a factor of $4x^3-21x^2+29x-6$

14 a)

Let $f(x)=2x^3-11x^2-9x+18$

$f(6)=432-396-54+18=0$

$\therefore x-6$ is a factor of $2x^3-11x^2-9x+18$

$2x^3-11x^2-9x+18=(x-6)(ax^2+bx+c)$

	$2x^2$	$+x$	-3
x	$2x^3$	$+x^2$	$-3x$
-6	$-12x^2$	$-6x$	$+18$

$2x^3-11x^2-9x+18=(x-6)(2x^2+x-3)$

$=(x-6)(x-1)(2x+3)$

14 b)

Let $f(x)=6x^3+19x^2+8x-5$

$f(-\frac{5}{2})=-\frac{375}{4}+\frac{475}{4}-20-5=0$

$\therefore 2x+5$ is a factor of $6x^3+19x^2+8x-5$

$6x^3+19x^2+8x-5=(2x+5)(ax^2+bx+c)$

	$3x^2$	$+2x$	-1
$2x$	$6x^3$	$+4x^2$	$-2x$
$+5$	$+15x^2$	$+10x$	-5

$6x^3+19x^2+8x-5=(2x+5)(3x^2+2x-1)$

$=(2x+5)(3x-1)(x+1)$

15)

Let $f(x)=(x+3)(x^2-kx+4)-20$

$f(-1)=0\Rightarrow(-1+3)(1+k+4)-20=0$

$\Rightarrow 2(k+5)-20=0\Rightarrow 2k+10-20=0\Rightarrow k=5$

16 a)

Let $f(x)=2x^3+7x^2-px-18$

$f(-3)=0\Rightarrow -54+63+3p-18=0\Rightarrow p=3$

16 b)

$2x^3+7x^2-3x-18=(x+3)(ax^2+bx+c)$

	$2x^2$	$+x$	-6
x	$2x^3$	$+x^2$	$-6x$
$+3$	$+6x^2$	$+3x$	-18

$2x^3+7x^2-3x-18=(x+3)(2x^2+x-6)$

$=(x+3)(2x-3)(x+2)$

17 a)

Let $f(x) = 32x^3 + px^2 + 219x + 45$

$f(-\frac{15}{8}) = 0 \Rightarrow -\frac{3375}{16} + \frac{225}{64}p - \frac{3285}{8} + 45 = 0$

$\Rightarrow p = 164$

17 b)

$32x^3 + 164x^2 + 219x + 45$

$= (8x + 15)(ax^2 + bx + c)$

	$4x^2$	$+13x$	$+3$
$8x$	$32x^3$	$+104x^2$	$+24x$
$+15$	$+60x^2$	$+195x$	$+45$

$32x^3 + 164x^2 + 219x + 45 = (8x + 15)(4x^2 + 13x + 3)$

$= (8x + 15)(4x + 1)(x + 3)$

18 a)

$f(-1) = -2 + 11 + p - 2p = 0 \Rightarrow p = 9$

18 b)

$2x^3 + 11x^2 - 9x - 18 = (x + 1)(ax^2 + bx + c)$

	$2x^2$	$+9x$	-18
x	$2x^3$	$+9x^2$	$-18x$
$+1$	$+2x^2$	$+9x$	-18

$2x^3 + 11x^2 - 9x - 18 = (x + 1)(2x^2 + 9x - 18)$

$= (x + 1)(2x - 3)(x + 6)$

19 a)

$f(-2) = -8p - 20 + 46 - q = 0 \Rightarrow -8p - q = -26$

$f(5) = 125p - 125 - 115 - q = 0$

$\Rightarrow 125p - q = 240$

$p = 2, q = 10$

19 b)

$(x + 2)(x - 5) = x^2 - 3x - 10$

	$2x$	$+1$
x^2	$2x^3$	$+x^2$
$-3x$	$-6x^2$	$-3x$
-10	$-20x$	-10

$2x^3 - 5x^2 - 23x - 10 = (x + 2)(x - 5)(2x + 1)$

20)

$f(-3) = 81p - 27q + 9r + 111 + 15 = 0$

$f(1) = p + q + r - 37 + 15 = 0$

$f(\frac{1}{2}) = \frac{1}{16}p + \frac{1}{8}q + \frac{1}{4}r - \frac{37}{2} + 15 = 0$

$81p - 27q + 9r = -126$

$p + q + r = 22$

$\frac{1}{16}p + \frac{1}{8}q + \frac{1}{4}r = \frac{7}{2}$

$p = 2, q = 13, r = 7$

21 a)

$f(-9) = 19683 - 43011 + 81p - 3321 - 162 = 0$

$\Rightarrow p = 331$

21 b)

$3x^4 + 59x^3 + 331x^2 + 369x - 162$

$= (x + 9)(\text{cubic})$

	$3x^3$	$+32x^2$	$+43x$	-18
x	$3x^4$	$+32x^3$	$+43x^2$	$-18x$
$+9$	$+27x^3$	$288x^2$	$+387x$	$+162$

$3x^4 + 59x^3 + 331x^2 + 369x - 162$

$= (x + 9)(3x^3 + 32x^2 + 43x - 18)$

Test other points:

$f(-2) = 48 - 472 + 1324 - 738 - 162 = 0$

$\therefore (x + 2)$ is a factor

	$3x^2$	$+26x$	-9
x	$3x^3$	$+26x^2$	$-9x$
$+2$	$+6x^2$	$+52x$	-18

$3x^4 + 59x^3 + 331x^2 + 369x - 162$

$= (x + 9)(x + 2)(3x^2 + 26x - 9)$

$= (x + 9)(x + 2)(3x - 1)(x + 9)$

22)

Let $f(x) = x^2 + bx + c$ and $g(x) = x^2 + px + q$

$f(1) = 1 + b + c = 0$

$g(-2) = 4 - 2p + q = 0$

$5 + b + c - 2p + q = 0$

23)

$135k^3 + 270k^2 + 165k + 30$

$= 5(27k^3 + 54k^2 + 33k + 6)$

Let $f(k) = 27k^3 + 54k^2 + 33k + 6$

$f(-\frac{1}{3}) = -1 + 6 - 11 + 6 = 0$

$\therefore (3k + 1)$ is a factor

	$9k^2$	$+15k$	$+6$
$3k$	$27k^3$	$+45k^2$	$+18k$
$+1$	$+9k^2$	$+15k$	$+6$

$135k^3 + 270k^2 + 165k + 30$

$= 5(3k + 1)(9k^2 + 15k + 6)$

$= 5(3k + 1)(3k + 2)(3k + 3)$

5×3 consecutive numbers

3 consecutive numbers are always a multiple of 6

$\therefore 135k^3 + 270k^2 + 165k + 30$ is always a multiple of 30

24 a)

$f(2) = 8 + 8 + 4 - 20 = 0$

$\therefore (x - 2)$ is a factor of $f(x)$

24 b)

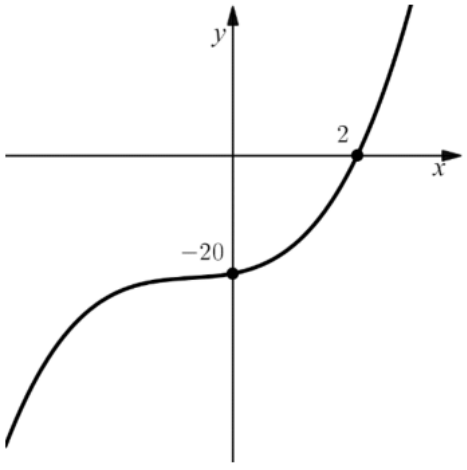
	x^2	$+4x$	$+10$
x	x^3	$+4x^2$	$+10x$
-2	$-2x^2$	$-8x$	-20

$x^3 + 2x^2 + 2x - 20 = (x - 2)(x^2 + 4x + 10)$

Discriminant of $x^2 + 4x + 10$ has discriminant $16 - 4 \times 1 \times 10 = -24 < 0$ so only solution to $f(x) = 0$ is $x = 2$

So the graph of $f(x)$ only intersects the x-axis once at $(2, 0)$

24 c)



25 a)

$f(-3) = -27 - 63 + 3p + 75 = 0 \Rightarrow p = 5$

25 b)

$f(x) = x^3 - 7x^2 - 5x + 75$

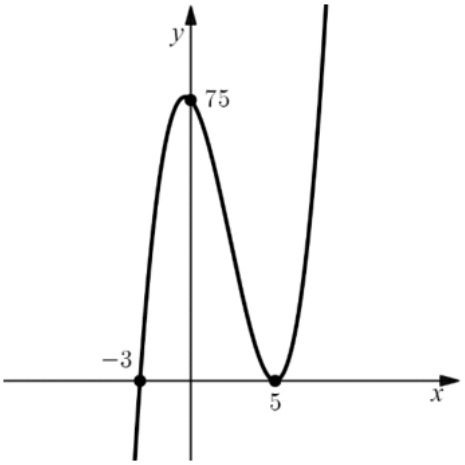
	x^2	$-10x$	$+25$
x	x^3	$-10x^2$	$+25x$
$+3$	$+3x^2$	$-30x$	$+75$

$x^3 - 7x^2 - 5x + 75 = (x+3)(x^2 - 10x + 25)$

$= (x+3)(x-5)^2$

$f(x) = 0$ when $x = -3, x = 5$

25 c)



26)

When $x = 0, y = 3k$

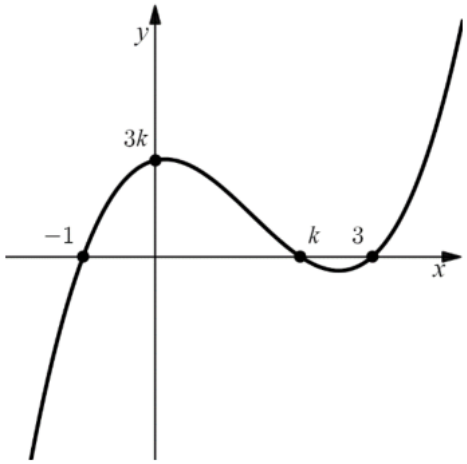
$(x+1)(x-3) = (x^2 - 2x - 3)$

	x	$-k$
x^2	x^3	$-kx^2$
$-2x$	$-2x^2$	$+2kx$
-3	$-3x$	$3k$

$x^3 - (2+k)x^2 + (2k-3)x + 3k$

$= (x+1)(x-3)(x-k)$

When $y = 0, x = -1, 3, x = k$



27 a)

$f(4) = 64 - 16(2k+4) + 4(k^2 + 8k) - 4k^2$

$= 64 - 32k - 64 + 4k^2 + 32k - 4k^2 = 0$

$\therefore (x-4)$ is a factor of $f(x)$ for all k

27 b)

	x^2	$-2kx$	$+k^2$
x	x^3	$-2kx^2$	$+k^2x$
-4	$-4x^2$	$+8kx$	$-4k^2$

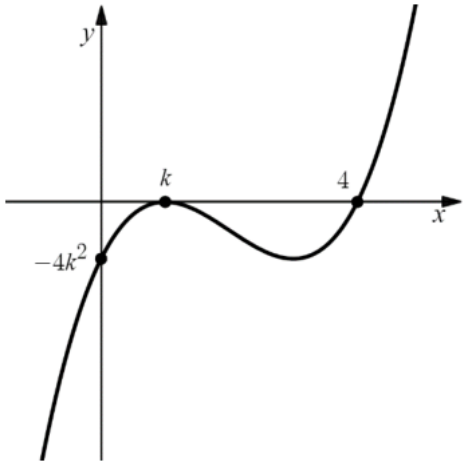
$x^3 - (2k+4)x^2 + (k^2+8k)x - 4k^2$

$= (x-4)(x^2 - 2kx + k^2)$

$x = \frac{2k \pm \sqrt{4k^2 - 4k^2}}{2} = \frac{2k}{2} = k$

$x^3 - (2k+4)x^2 + (k^2+8k)x - 4k^2$

$= (x-4)(x-k)^2$



28 a)

	x^2	$+0x$	$+0$	
x	x^3	$+0x^2$	$+0x$	$+5$
-1	$-x^2$	$-0x$	$+0$	

$(x^3 - x^2 + 5) \div (x-1) = x^2$ remainder 5

$\frac{x^3 - x^2 + 5}{x-1} = x^2 + \frac{5}{x-1}$

28 b)

At $x = 0, y = -5$

Asymptote at $x = -1$

As $x \to \pm\infty, \frac{x^3 - x^2 + 5}{x - 1} \to x^2$

